Looping Constructs in C#.Net

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Overview

For Loop

While Loop

Do While Loop

For Each Loop

Introduction to Looping Constructs in C#.Net

- **The For Loop**
- The "for" loop is a common looping construct in C# used to iterate a specific number of times.
- It provides a compact and structured way to execute a block of code repeatedly.
- The loop consists of an initialization statement, a condition, and an iteration statement, all within a single line.

Syntax:

```
for([Initialization] ; [Condition] ; [Updation])
{
        [Expression/Statement];
}
```

Example:

```
int n = 5;
int factorial = 1;

for (int i = 1; i <= n; i++)
{
    factorial *= i;
}

Console.WriteLine("Factorial of " + n + " is: " + factorial);
```

- **The while Loop**

- The "while" loop is a fundamental loop construct in C# that repeatedly executes a block of code as long as a specified condition is true.
- The loop checks the condition before executing the loop body, and if the condition evaluates to true, the loop body is executed.
- If the condition is initially false, the loop body is skipped entirely.

Syntax:

```
[Initialization]
While([Condition])
{
        [Expression/Statement];
        Updation;
}
```

Example:

```
int number = 12345;
int sum = 0;

while (number > 0)
{
   int digit = number % 10;
   sum += digit;
   number /= 10;
}

Console.WriteLine("Sum of digits: " + sum);
```

- **The do while Loop**

- The "do-while" loop is a type of loop in C# that executes a block of code repeatedly until a specified condition is no longer true.
- Unlike the "while" loop, which checks the condition before the loop body, the "do-while" loop checks the condition after executing the loop body.
- This guarantees that the loop body will execute at least once, even if the condition is initially false.

Syntax:

```
[Initialization]
do
{
        [Expression/Statement];
        Updation;
}while([Condition])
```

Example:

```
int number = 12345;
int sum = 0;

do
{
    int digit = number % 10;
    sum += digit;
    number /= 10;
}while(number > 0);

Console.WriteLine("Sum of digits: " + sum);
```

The foreach Loop

- The foreach loop in C# is used to iterate over elements in a collection or an array.
- It simplifies the process of iterating over each element without the need for explicitly managing an index or specifying the loop range.
- The foreach loop automatically iterates over each element in the collection until all elements have been processed.

Syntax:

```
foreach([Type] [Identifier] in [collection])
{
            [Expression/Statement];
}
```

Example:

```
foreach (elementType element in collection)
{
   // Code to be executed for each element
}
```

References

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